Pol Cardona Rubio

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Pragmatic engineer, finishing an applied PhD in optimization-based control at IRI (CSIC–UPC), on green-hydrogen refuelling stations: modelling the plant and designing controllers that solve real operational problems.



Work Experience

- **PhD Candidate**
  Institut de Robòtica i Informàtica Industrial (IRI), CSIC-UPC.
  February 2023 – Present (defending 2026) · 3 years 8 months
  Focus: optimization-based supervisory control of on-site renewable Hydrogen Refuelling Stations (HRSs) for the transport sector, grounded in the Zaragoza HRS pilot of the PTI-TRANSENER+ project.
  Achievements:
  - Built a Mixed Logical Dynamical (MLD) model of the Zaragoza HRS, capturing multi-pressure storage, the event-triggered arrival of Fuel-Cell Electric Vehicles (FCEVs) and the discrete equipment states in a single linear-inequality framework, solvable as a Mixed-Integer Linear or Quadratic Program (MILP/MIQP) inside an online receding-horizon controller *(Applied Energy 2025)*.
  - Designed and tested linear Model Predictive Control (MPC) and chance-constrained MPC schemes for the same plant. Linear MPC outperformed rule-based dispatch across all scenarios of the sensitivity sweep *(IJHE 2023)*; the chance-constrained variant handled the uncertain initial hydrogen mass at FCEV arrival *(EHEC 2024 poster)*.
  - Researching hydrogen storage readiness for vehicle refuelling *(manuscript in preparation, 2026)*.
  - Co-supervised a master's thesis on a techno-economic analysis of a national-scale renewable HRS network for Spain *(IJHE 2026)*.
- **Research Support Engineer**
  Institut de Robòtica i Informàtica Industrial, CSIC-UPC.
  August 2021 – July 2025 · 3 years 11 months
  Project: PTI-TRANSENER+ [TRE2103000], Renewable Hydrogen Refuelling Station, with industry partners.
  `https://www.iri.upc.edu/project/show/264`
  Achievements:
  - Built two models of the Zaragoza HRS: a nonlinear one for simulation, and a linearized one for control design.
  - Wrote the rule-based control strategy and the day-ahead scheduling algorithm that picks the grid-connected hydrogen production hours, along with the demand and renewable-generation scenario sets used to evaluate them.
  - Software-in-the-Loop integration: ported the Simulink HRS model, the state machine and the day-ahead scheduler into LabVIEW on a CompactRIO for testing communication layers with external and auxiliary hardware.

- **Junior Product Manager**
  Groupe Atlantic. Renewable energy catalogue, Spain and Portugal.
  June 2019 – May 2021 · 1 year 11 months
  Achievements:
  - Co-developed a new kit product that combined photovoltaic and aerothermal energy, from supplier selection to sales team presentation and launch into the Iberian catalogue.
  - Ran technical and commercial market studies of the renewable-energy catalogue for Spain and Portugal.
  - Maintained and improved several databases covering product references, commercial benchmarking and annual sales targets.
  - Reformatted technical product data to meet individual clients' requirements and sector-wide product catalogues.
  - Delivered and revised technical documentation on aerothermal, photovoltaic and solar thermal systems for the upper-level management and sales team.

Projects & Leadership

- **UPC Vilanova E3-Team (MotoStudent Electric, EPSEVG–UPC)**
  Team Leader: Jul 2018–Sep 2021 · 3 years 2 months.
  Electronic Technician: Oct 2016–Jul 2018 · 1 year 9 months.
  Funding: UPC and sponsors.
  Achievements: built and tested e-powertrain subsystems; as Team Leader, helped rebuild the team from the ground up, prioritising internal knowledge transfer and lasting values of humility, practicality, ambition and efficiency; doubled membership and the per-edition budget through new sponsorships and increased UPC support; coordinated vehicle integration and track testing.

Publications & Work in Progress

Work in progress

- **Hydrogen storage readiness for refuelling stations**.
  Pol Cardona, Carlos Ocampo-Martinez, Maria Serra.
  Manuscript in preparation (2026).

Published

- **Techno-economic analysis of a renewable hydrogen refuelling station network: The case study of Spain**.
  B. Garcia Minguez, Pol Cardona, Luis Valiño, Carlos Ocampo-Martinez, L. Siemon, Maria Serra.
  International Journal of Hydrogen Energy (2026).
  DOI: https://doi.org/10.1016/j.ijhydene.2025.152325
- **Mixed logical dynamical modelling of renewable hydrogen refuelling stations for the design of optimization-based operational schemes**.
  Pol Cardona, Luis Valiño, Carlos Ocampo-Martinez, Maria Serra.
  Applied Energy, 393 (2025) 126037.
  DOI: https://doi.org/10.1016/j.apenergy.2025.126037
- **Model predictive control of an on-site green hydrogen production and refuelling station**.
  Pol Cardona, Ramon Costa-Castelló, Vicente Roda, Javier Carroquino, Luis Valiño, Maria Serra.
  International Journal of Hydrogen Energy (2023).
  DOI: https://doi.org/10.1016/j.ijhydene.2023.01.191

- **Modelling and operation strategy approaches for on-site Hydrogen Refuelling Stations**.
  Pol Cardona, Ramon Costa-Castelló, Vicente Roda, Javier Carroquino, Luis Valiño, Carlos Ocampo-Martinez, Maria Serra.
  International Journal of Hydrogen Energy (2023).
  DOI: https://doi.org/10.1016/j.ijhydene.2023.08.192

Conferences & Posters

- **12th European Fuel Cells and Hydrogen Piero Lunghi Conference (EFCH$_2$ 2026)**. Capri, Italy; 15–17 September 2026.
  Attended and delivered the oral presentation (ID 215): *Predictive operation of a hydrogen refuelling station with scheduled clients.*
  Authors: Pol Cardona, Carlos Ocampo-Martinez, Maria Serra Prat.
  Conference website | Official programme (presentation ID 215)
- **European Hydrogen Energy Conference (EHEC 2024)**. Bilbao Exhibition Centre (BEC), Bilbao, Spain; 6–8 March 2024.
  Poster P54 (ID204): *Chance-constrained model predictive control for green and grid-connected hydrogen refueling stations.*
  Authors: Pol Cardona, Luis Valiño, Carlos Ocampo-Martinez, Maria Serra.
- **II Simposio Conjunto Grupos Temáticos CEA: Ingeniería de Control y Modelado, Simulación y Optimización**. Madrid, Spain; April 2023.
  Talk: *Control óptimo de estaciones de repostaje de hidrógeno para el sector transporte.*
  Authors: P. Cardona, M. Serra, C. Ocampo-Martinez.
- **VIII Symposium on Hydrogen, Fuel Cells and Advanced Batteries (Hyceltec 2022)**. Buenos Aires, Argentina; 11–14 July 2022.
  Proceedings: *State Machine for efficient operation of an on-site green production hydrogen refueling station (P-8H-V)*; p. 156. Source
  Authors: P. Cardona, E. Recarte, V. Roda, L. Valiño, M. Serra.

Education & Further Training

- **Universitat Politècnica de Catalunya (UPC)**
  Master's in Industrial Engineering
  Start: *September 2020*, Completion: *February 2023*
  Average grade: 7.26/10.
- **Universitat Politècnica de Catalunya (UPC)**
  Bachelor's in Electronic Industrial and Automation Engineering
  Start: *September 2016*, Completion: *July 2020*
  Average grade: 7.85/10.
- **Hybrid Control Systems – EECI IGSC 2025 (M14)**
  2–6 June 2025 · 21-hour course.
  Instructor: Ricardo Sanfelice (UC Santa Cruz). UCLouvain, Louvain-la-Neuve, Belgium.
  Official programme

Skills & Languages

- **Optimization & Control:** MPC, Hybrid MPC, Mixed Logical Dynamical modelling, MILP/MIQP/NLP, chance-constrained, lexicographic optimization, day-ahead scheduling.
- **Modelling & Energy Systems:** Hydrogen Refuelling Station components (electrolyzer, compressor, multi-pressure storage, chiller, dispenser), PV/wind, microgrids; control-oriented and simulation-oriented modelling.
- **Tools & Solvers:** MATLAB/Simulink/Stateflow, Python, CasADi, YALMIP, HYSDEL, Pyomo/AMPL; Gurobi, CPLEX, Octeract; LaTeX.
- **Data:** Multi-scenario design, time-series analysis and visualization.
- **Languages:** Catalan (native), Spanish (native), English (fluent).